3.7 Hardware setup:

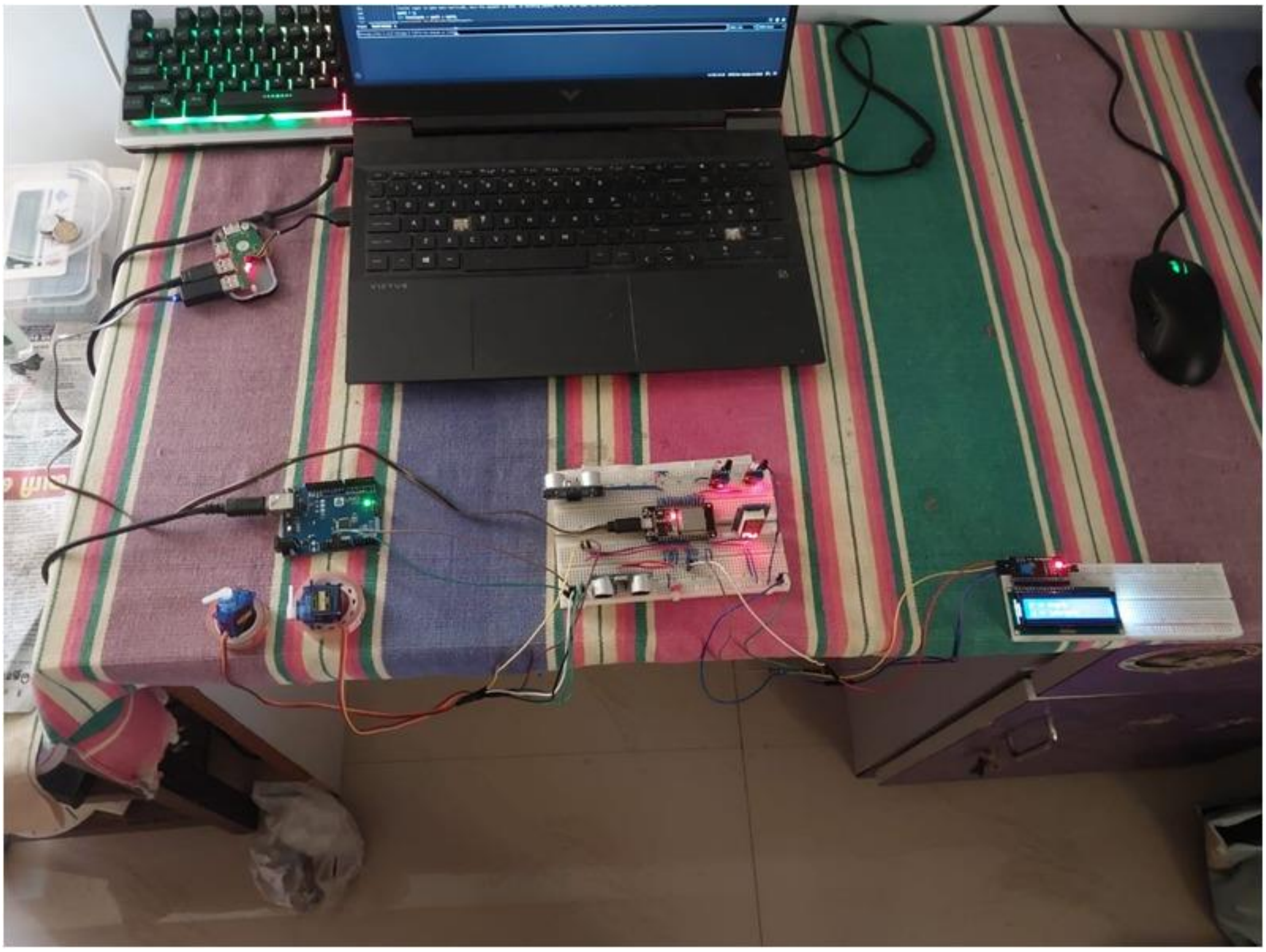
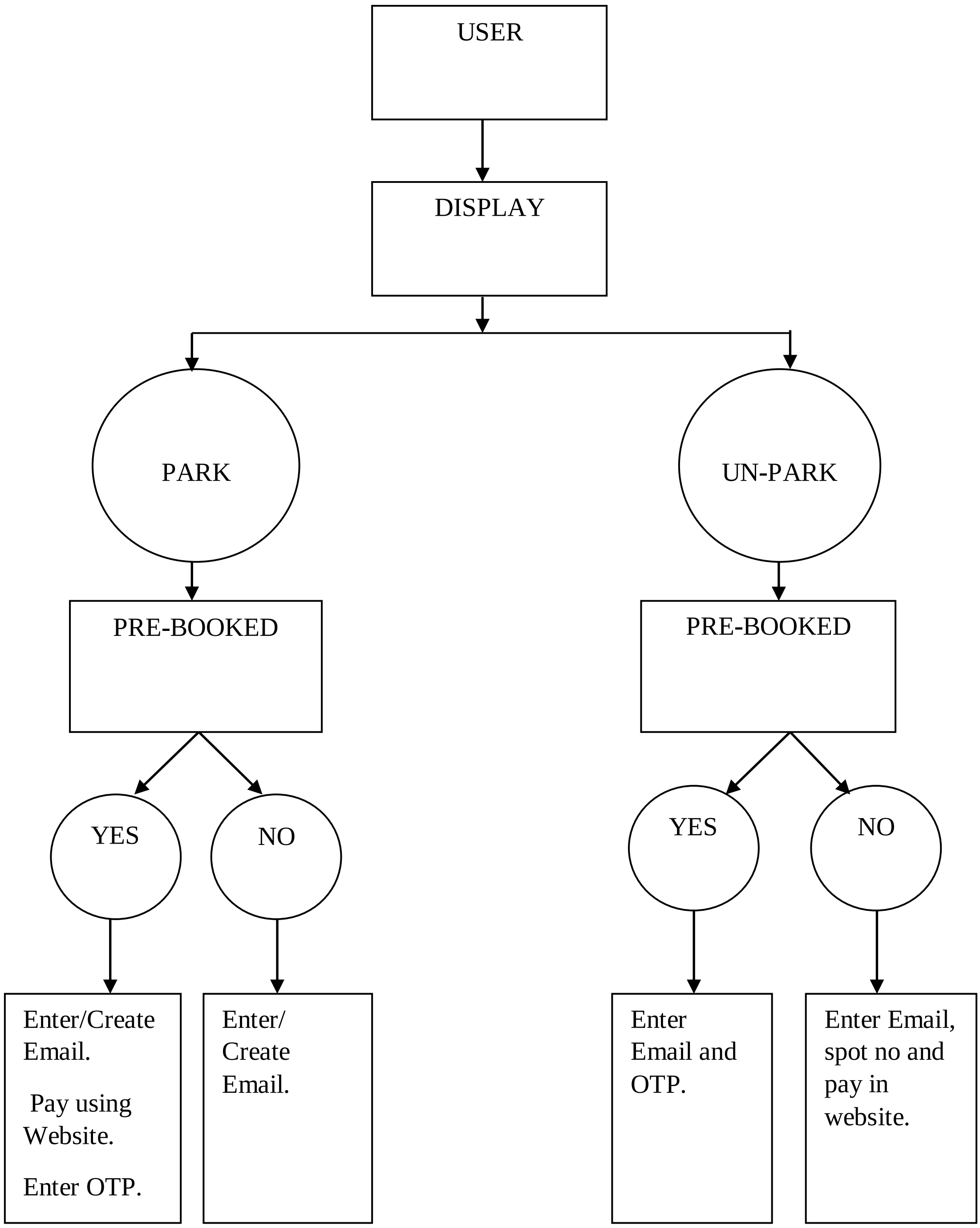


Figure 3.7: MAIN SETUP OF THE PROTOTYPE

3.8 Proposed Methodology:



3.8.1 Flowchart of the working system

3.8.2 Workflow:

The proposed project works in a gist like the below:

A person, may not be a user yet, comes to the parking lot and sees the **DISPLAY.** When prompted the person would be asked a question.

Park or un-park (p or u)?

- If user presses 'p', the person would then see a question. **“Pre-booked”**, **Yes or No**?
  - If user presses **“y”**, user is asked for 'registered email' and **“OTP”**
    - If either is wrong. The IoT system would return back to the main prompt.
    - If entered email doesn't exist in the user database, the user is prompted to register an email from the complementary website for the proposed smart parking system. And the IoT system would return back to the main prompt.
    - If the entered details are correct, user is let in by opening up the main barricade and the pre-booked spot is opened up automatically for the user to park.
  - If user presses **“n”**, user is asked for 'registered email'
    - If entered email doesn't exist in the user database the user is prompted to register an email from the complementary website for the proposed smart parking system.And the IoT system would return back to the main prompt.
    - Else, the main barricade is opened up and the user can drive up-to any FREE parking spot and its corresponding barricade would open up. The email is then tracked for the time for which the user is parked.

- If user presses **“u”**, the person would then see a question. **“Pre-booked”**, yes or no?
  - If user presses **“y”**, user is asked for 'registered email' and 'OTP'.
    - If either combination is wrong, the IoT system would return back to the main prompt.

        - If entered email doesn't exist in the user database, the IoT system would return back to the main prompt.
        - If the entered details are correct, user is let out by opening up the main barricade and the pre-booked spot is opened up automatically for the user to un-park.
    - If user presses **“n”**, user is asked for the 'registered email'.
        - If the entered email doesn't exist in the user database, the IoT system would return back to the main prompt.
        - Else, the user is prompted to pay his/her fee due in the supplementary website of the IoT system.
    - The main barricade stays closed until the user logs into the website using the registered email, that was used whilst parking and pays his/her due using a simple**“pay”**.
    - After the user PAYS, the main barricade opens and user can leave.

3.9 Website

- A website is a crucial software part for the proposed IoT system, as it serves the purpose of the user interacting remotely with the IoT system without any need for internal details of the parking lot.
- The website is hosted using the traditional components like HTML, CSS and JavaScript.
- The JavaScript file, attached to the HTML file, would be in direct knots with the whole backend part of the IoT system.
- It queries for various information, updates various information and also gets the user database details from the backend.

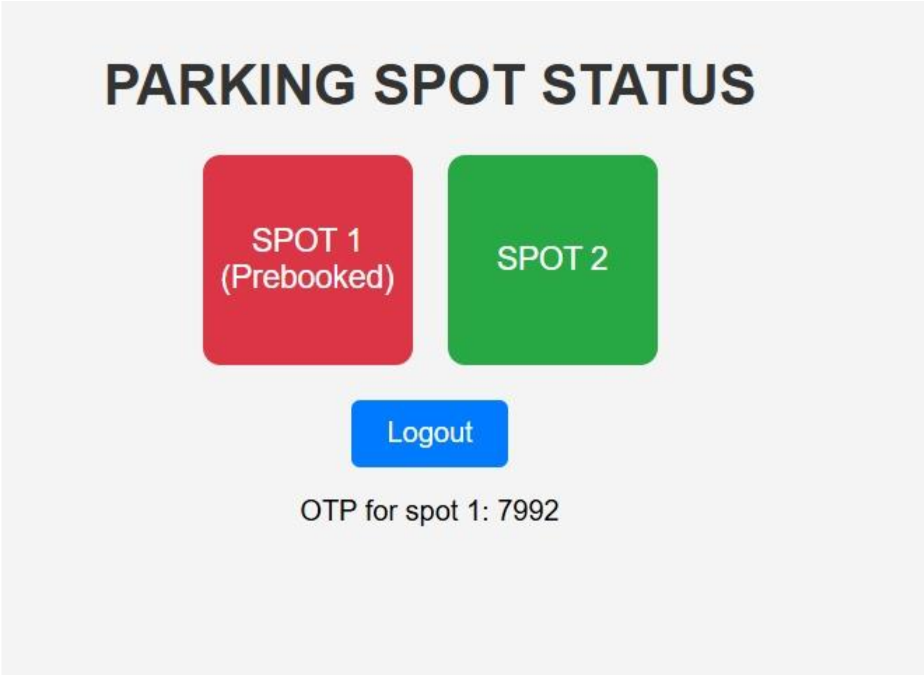


Figure 4.1.3: Non pre-booked un-parking spot 2

Figure 4.1.3 shows the process of un-parking from spot 2, which was not pre-booked. It represents a standard exit after temporary parking without a reservation.

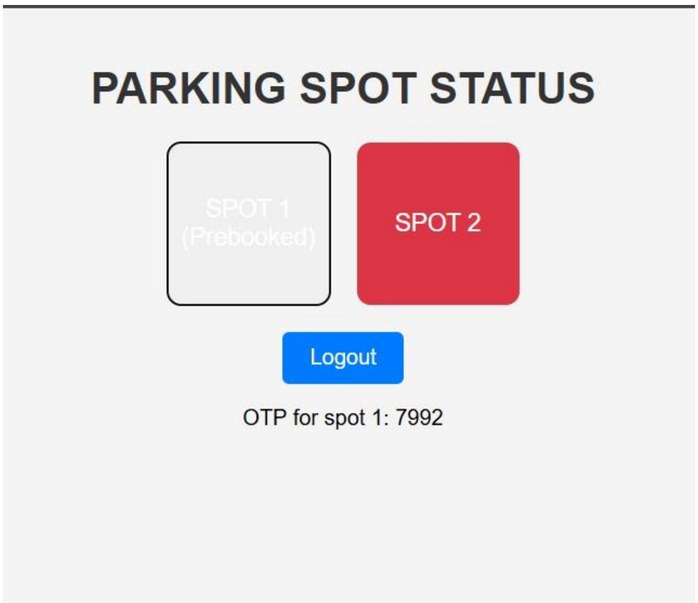


Figure 4.1.4: Pre-booked parking slot 1

Figure 4.1.4 illustrates parking slot 1 being used after a successful pre-booking. This confirms that the user has reserved the slot in advance before parking.

The above buttons represent car parking spots 1 and 2 (IR/Ultrasonic sensors 1 and 2)

Spot 1 is a button that gets spot data from backend using REST API

Spot 2 is a button that gets spot data from backend using REST API

Backend is connected to the frontend (website), which sends information about the spot status, obtained from the IoT hardware.

IoT hardware is in turn, connected to the backend.

4.2. Hardware Results

```
P = Park
U = Unpark
u was pressed

Response: {"spot1state":0,"spot2state":0,"spot1otp":null,"spot2otp":null}
Spot 1 State: 0
Spot 2 State: 0
Spot 1 OTP: 0
Spot 2 OTP: 0
Unparking mode!
Pre-booked?
y = yes
no = no
y was pressed

Enter spot no:
Enter OTP from your account:
Take out your vehicle!
```

Figure 4.2.1: Pre-booked un-parking spot ½ correct email and OTP combination

Figure 4.2.1 shows the process of unparking from a pre-booked spot (either 1 or 2) using the correct email and OTP combination. This verifies authorized access and ensures secure un-parking.

```
P = Park
U = Unpark
p was pressed

Response: {"spot1state":0,"spot2state":0,"spot1otp":null,"spot2otp":null}
Spot 1 State: 0
Spot 2 State: 0
Spot 1 OTP: 0
Spot 2 OTP: 0
Parking mode!
Pre-booked?
y = yes
no = no
y was pressed

Enter spot no:
Spot 2 is not prebooked!
```

Figure 4.2.2: Pre-booked parking spot 2Late arrival

Figure 4.2.2 illustrates a pre-booked parking spot 2 being used after a late arrival. Despite the delay, the spot remains reserved for the user, reflecting the flexibility of the pre-booking system.

```
P = Park
U = Unpark
p was pressed

Response: {"spot1state":0,"spot2state":1,"spot1otp":null,"spot2otp":1912}
Spot 1 State: 0
Spot 2 State: 1
Spot 1 OTP: 0
Spot 2 OTP: 1912
Parking mode!
Pre-booked?
y = yes
no = no
y was pressed

Enter spot no:
Enter OTP from your account:
Enter OTP correctly!
```

Figure 4.2.3: Pre-booked parking spot 2wrong OTP

Figure 4.2.3 shows the scenario where a pre-booked parking spot 2 is accessed using an incorrect OTP. This prevents unauthorized access, ensuring only valid users can park in the reserved spot.

```
P = Park
U = Unpark
u was pressed

Response: {"spot1state":1,"spot2state":0,"spot1otp":9055,"spot2otp":null}
Spot 1 State: 1
Spot 2 State: 0
Spot 1 OTP: 9055
Spot 2 OTP: 0
Unparking mode!
Pre-booked?
y = yes
no = no
n was pressed

Response: {"spot1state":1,"spot2state":0,"spot1otp":9055,"spot2otp":null}
Spot 1 State: 1
Spot 2 State: 0
Spot 1 OTP: 9055
Spot 2 OTP: 0
Enter spot no:
2 was pressed
Take out your vehicle
```

Figure 4.2.4: Non Pre-booked un-parking spot 2

Figure 4.2.4 demonstrates the process of un-parking from spot 2, which was not pre-booked. This reflects a typical exit from a parking spot without any prior reservation.

```
P = Park
U = Unpark
u was pressed

Response: {"spot1state":1,"spot2state":0,"spot1otp":9055,"spot2otp":null}
Spot 1 State: 1
Spot 2 State: 0
Spot 1 OTP: 9055
Spot 2 OTP: 0
Unparking mode!
Pre-booked?
y = yes
no = no
n was pressed

Response: {"spot1state":1,"spot2state":0,"spot1otp":9055,"spot2otp":null}
Spot 1 State: 1
Spot 2 State: 0
Spot 1 OTP: 9055
Spot 2 OTP: 0
Enter spot no:
2 was pressed
No vechicle in2
```

Figure 4.2.5: Non Pre-booked un-parking of empty spot (no vehicle)

Figure 4.2.5 shows the process of attempting to un-park from an empty spot (no vehicle present) that was not pre-booked. This scenario may indicate an error or unnecessary action, as there is no vehicle occupying the spot.

```
Pre-booked?
y = yes
no = no
n was pressed

Response: {"spot1state":0,"spot2state":0,"spot1otp":null,"spot2otp":null}
Spot 1 State: 0
Spot 2 State: 0
Spot 1 OTP: 0
Spot 2 OTP: 0
Enter the email you have linked your account to:
Entered email: gokula@gmail.com

Email exists!
Response: {"spot1state":0,"spot2state":0,"spot1otp":null,"spot2otp":null}
Spot 1 State: 0
Spot 2 State: 0
Spot 1 OTP: 0
Spot 2 OTP: 0


Park in free spots!
Spot 2 is being taken
```

Figure 4.2.6: Non pre-booked parking spot 2correct email

Figure 4.2.6 depicts the scenario where parking spot 2 is accessed without a pre-booking, but the user enters the correct email. This could indicate a verification step for parking access, even without a reservation.

```
Parking mode!
Pre-booked?
y = yes
no = no
n was pressed

Response: {"spot1state":0,"spot2state":0,"spot1otp":null,"spot2otp":null}
Spot 1 State: 0
Spot 2 State: 0
Spot 1 OTP: 0
Spot 2 OTP: 0
Enter the email you have linked your account to:
Entered email: asdijoads

No such Email exists!

You need to be signed up in our website, to proceed!
```

Figure 4.2.7: Non pre-booked parking spot½ wrong email

Figure 4.2.7 shows the scenario where a non-pre-booked parking spot (either 1 or 2) is accessed with an incorrect email. This prevents access to the parking spot, ensuring only authorized users can park.